Secure Boot Design with Hardware Root of Trust as a core for EFR32MG26

Description:

Participants must engineer a Hardened Secure Boot architecture for the EFR32MG26, utilizing Secure Vault High and the immutable Root of Trust Secure Loader (RTSL) to guarantee firmware authenticity. To thwart physical perturbation, the design must replace standard boolean checks with glitch-resistant control flow, employing layered token verification and TRNG-based random jitter to desynchronize fault injection timing. Active environment tracking using ACMP/IADC peripherals is required to trigger preemptive anti-tamper responses against voltage or temperature anomalies. Architecture must enforce anti-rollback via One-Time Programmable (OTP) counters to preclude downgrade attacks, while facilitating secure debug for authorized field diagnostics. Furthermore, the solution must evolve into a measured boot system that generates signed health reports for remote attestation and forensics. Critical boot logic must be isolated within the TrustZone Secure World, and key wrapping via the Physically Unclonable Function (PUF) must be implemented to protect against memory dump extraction.

Relevance:

1. Defeating Physical Attacks: Mitigates single-cycle voltage glitches that bypass standard checks by requiring complex, multi-fault injection sequences.
2. Zero Trust Foundation: Establishes a trust chain where devices prove integrity via cryptographic health reports rather than just booting silently.
3. Isolation of Trust: TrustZone placement prevents application-layer vulnerabilities (e.g., in BLE stacks) from compromising root keys or the bootloader.

Business Case:

1. **National Infrastructure:** Prevents "kill-switch" attacks on critical grids (smart meters) by state-sponsored actors.
2. **IP Preservation:** Blocks competitors from dumping decrypted firmware from RAM, securing RD investments.
3. **Compliance:** Essential for FIPS 140-3 Level 3 and Common Criteria EAL4+ certification required for government contracts.
4. **Strategic Autonomy:** Strengthens "Atmanirbhar Bharat" by reducing reliance on opaque, black-box foreign security solutions.

Key Deliverables:

1. **Hardened Bootloader:** Binary implementing non-binary token verification and random jitter logic.
2. **Tamper Configuration:** C-code initializing EFR32 ACMP/IADC monitors for preemptive fault detection.
3. **Attestation Schema:** JSON/CBOR format for the signed boot health/telemetry report.
4. **Validation Report:** Evidence of resilience against Chip Whisperer voltage glitching attempts.

Resources & Notes:

1. **Silicon Labs AN1218:** Series 2 Secure Boot with RTSL.
2. **Silicon Labs AN1247:** Anti-Tamper Protection Configuration.
3. **NIST SP 800-140:** FIPS 140-3 Security Requirements.
4. **AIS 189:** Automotive Cyber Security Standards (India).

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